Tertiary Infotech Academy Pte Ltd
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LESSON PLAN

For

Claude AI for Digital Marketing

Course Code: C141

Conducted by

Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
3.0	12 Oct 2026	Rebuilt as Claude AI for Digital Marketing from the parent course's v10.0 courseware: the same three topics and 15 hands-on labs, in which learners build an AI marketing team for Horizon Wealth Planning in Claude Code and run campaigns with it, over two days of 9:30am – 5:30pm (7.5 instructional hours per day) with one 30-minute lunch break. Supersedes the earlier C141 courseware (v1.0–v2.0), published under the course's previous title.	Dr. Alfred Ang

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TABLE OF CONTENTS

Course Overview **4**

Learning Outcomes **4**

Daily Schedule **4**

Day 1 — Form, Assess and Integrate the Team 4

Day 2 — Test, Troubleshoot and Version 5

Topic-by-topic Breakdown **6**

Topic 1 — Claude Cowork Components, Compatibility and Integration · Slides 26–73 6

Lab 1 — Build Your Marketing Agent Team · Slides 32–36 6

Lab 2 — Form the Team: Instructions for Each Agent · Slides 39–44 7

Lab 3 — Give Each Agent Its Skills · Slides 47–52 7

Lab 4 — Connect the Tools and Check Compatibility · Slides 57–62 7

Lab 5 — Campaign 1: Research to Storyboard, with Subagents · Slides 66–72 7

Topic 2 — Testing Marketing Integrations and Troubleshooting Errors · Slides 74–138 8

Lab 6 — Growth Analyst: Personas from Evidence · Slides 79–84 8

Lab 7 — Growth Analyst and Strategist: Cadence and Calendar · Slides 86–91 8

Lab 8 — Content Creator: A Quick Blog Article · Slides 93–97 8

Lab 9 — The Lead Magnet: a Checklist and Its Welcome Emails · Slides 100–107 9

Lab 10 — Campaign 2: Social Media Posts (Demo) · Slides 117–122 9

Lab 11 — Campaign 3: Newsletter and a Landing Page · Slides 125–130 9

Lab 12 — Campaign 4: The YouTube Explainer · Slides 132–137 10

Topic 3 — Versioning, Change Proposals and Integration Best Practices · Slides 139–176 10

Lab 13 — Campaign 5: Always On — the Weekly Growth Report · Slides 144–149 10

Lab 14 — Measure, Version and Write the Team Playbook · Slides 154–160 10

Lab 15 — Agent Teams: A Strategy Debate and a QA Loop · Slides 164–170 11

Resources Required **11**

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Course Overview

A two-day, hands-on course in system integration for digital marketing with Claude Cowork and Claude Code. Learners take one business — Horizon Wealth Planning, an independent financial-planning firm in Singapore with a website and no marketing team — and build it an AI marketing team: a Marketing Team Lead and five specialists (Growth Strategist, Content Creator, Creative Designer, Website Designer, Growth Analyst), each with instructions, skills and connectors. They assess the compatibility of every component and interface and integrate them (Topic 1); test each integration through real campaign work and troubleshoot the errors (Topic 2); and version the team, propose changes and apply integration best practices (Topic 3), with a person approving every external piece. The course uses Claude Code subagents and agent teams, Claude connectors (MCP) and scheduled tasks, and publishes pages as Claude artifacts.

Course code C141 · 2 days · 15 hours (9:30am – 5:30pm, 7.5 instructional hours per day) · 176-slide deck v3.0 · 15 labs, each in its own folder.

Learning Outcomes

By the end of this course, participants will be able to:

- **LO1.** Conduct a compatibility assessment of Claude Cowork interfaces and use essential tools to integrate selected marketing components.
- **LO2.** Test the integrated marketing components and interfaces, identify integration errors and conduct basic troubleshooting.
- **LO3.** Propose changes to the integration plan from observed results, through versioning, and apply integration best practices.

Daily Schedule

Both days run 9:30 AM – 5:30 PM: 7.5 instructional hours with one 30-minute lunch break and no separate tea breaks. The Slides column gives the deck pages for each teaching block.

Day 1 — Form, Assess and Integrate the Team

Time	Duration	Topic / Activity	Method	Slides
9:30 AM–10:10 AM	40 min	Welcome, trainer, ground rules, learning outcomes; overview — the Claude suite, Claude Cowork, Claude Code, plans and pricing, AI 2023–2026, the harness, the agentic loop, agentic AI vs AI agents, what an agent is made of; the Horizon scenario and your AI marketing team, lesson plan	Presentation, discussion	1–25
10:10 AM–10:30 AM	20 min	Topic 1 — why a team of agents; the systems behind the team; components and interfaces; getting the tools ready	Presentation, live demo	26–31
10:30 AM–11:00 AM	30 min	Lab 1 — Build Your Marketing Agent Team	Hands-on lab	32–36
11:00 AM–11:10 AM	10 min	Agent instructions: job, inputs, outputs, limits	Presentation, live demo	37–38

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11:10 AM–11:50 AM	40 min	Lab 2 — Form the Team: Instructions for Each Agent	Hands-on lab	39–44
11:50 AM–12:00 PM	10 min	Skills: twenty procedures, owned by role	Presentation	45–46
12:00 PM–12:45 PM	45 min	Lab 3 — Give Each Agent Its Skills	Hands-on lab	47–52
12:45 PM–1:15 PM	30 min	Lunch	—	—
1:15 PM–1:30 PM	15 min	Connectors and least privilege; assessing compatibility; the compatibility matrix	Presentation	53–56
1:30 PM–2:15 PM	45 min	Lab 4 — Connect the Tools and Check Compatibility	Hands-on lab	57–62
2:15 PM–2:25 PM	10 min	Campaign 1: how the integrated team works; the five-beat storyboard; regulated content	Presentation	63–65
2:25 PM–3:05 PM	40 min	Lab 5 — Campaign 1: Research to Storyboard, with Subagents	Hands-on lab	66–72
3:05 PM–3:10 PM	5 min	Topic 1 recap	Discussion	73
3:10 PM–3:20 PM	10 min	Topic 2 — integration tools and techniques; every campaign is an integration test; evidence before personas	Presentation	74–78
3:20 PM–4:00 PM	40 min	Lab 6 — Growth Analyst: Personas from Evidence	Hands-on lab	79–84
4:00 PM–4:05 PM	5 min	Finding the right frequency	Presentation	85
4:05 PM–4:40 PM	35 min	Lab 7 — Growth Analyst and Strategist: Cadence and Calendar	Hands-on lab	86–91
4:40 PM–4:45 PM	5 min	From storyboard to content: a quick blog in one prompt	Presentation	92
4:45 PM–5:25 PM	40 min	Lab 8 — Content Creator: A Quick Blog Article	Hands-on lab	93–97
5:25 PM–5:30 PM	5 min	Day 1 review and Q&A	Discussion	98

Day 2 — Test, Troubleshoot and Version

Time	Duration	Topic / Activity	Method	Slides
9:30 AM–9:35 AM	5 min	The lead magnet: content that earns a sign-up	Presentation	99
9:35 AM–10:25 AM	50 min	Lab 9 — The Lead Magnet: a Checklist and Its Welcome Emails	Hands-on lab	100–107
10:25 AM–10:45 AM	20 min	Subagents vs agent teams; human in the loop, enforced; protocols at each interface; the publishing APIs; signs of integration errors	Presentation, live demo	108–116
10:45 AM–11:45 AM	60 min	Lab 10 — Campaign 2: Social Media Posts (Demo)	Hands-on lab	117–122
11:45 AM–11:50 AM	5 min	Campaign 3: newsletter, landing page and consent	Presentation	123–124

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11:50 AM–12:50 PM	60 min	Lab 11 — Campaign 3: Newsletter and a Landing Page	Hands-on lab	125–130
12:50 PM–1:20 PM	30 min	Lunch	—	—
1:20 PM–1:25 PM	5 min	Campaign 4: the video team	Presentation	131
1:25 PM–2:10 PM	45 min	Lab 12 — Campaign 4: The YouTube Explainer	Hands-on lab	132–137
2:10 PM–2:15 PM	5 min	Topic 2 recap	Discussion	138
2:15 PM–2:35 PM	20 min	Topic 3 — troubleshooting methodologies; always on: from skill to schedule	Presentation, live demo	139–143
2:35 PM–3:20 PM	45 min	Lab 13 — Campaign 5: Always On — the Weekly Growth Report	Hands-on lab	144–149
3:20 PM–3:35 PM	15 min	Responsible AI; measuring what matters; versioning the team; a good change proposal	Presentation	150–153
3:35 PM–4:35 PM	60 min	Lab 14 — Measure, Version and Write the Team Playbook	Hands-on lab	154–160
4:35 PM–4:45 PM	10 min	Agent teams at work: subagents vs agent teams; a team that argues; the QA loop	Presentation	161–163
4:45 PM–5:20 PM	35 min	Lab 15 — Agent Teams: A Strategy Debate and a QA Loop	Hands-on lab	164–170
5:20 PM–5:30 PM	10 min	Course summary: integration best practices; Q&A; course feedback	Discussion	171–176

Topic-by-topic Breakdown

Topic 1 — Claude Cowork Components, Compatibility and Integration · Slides 26–73

- Why a team of agents
- The systems behind the team, their components and interfaces
- A home for the team; Horizon's website
- Form the team: instructions, skills, connectors
- Assess compatibility, integrate, then run Campaign 1

Lab 1 — Build Your Marketing Agent Team · Slides 32–36

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code
- **Time: 30 minutes (Day 1)**
- **Features practised:** CLAUDE.md as the team charter · your own copy of Horizon's website
- **Output:** horizon-studio/ with CLAUDE.md, the shared data, and your own copy of the site running at http://localhost:8080/
- **Evidence:** Claude shows horizon-studio as the chosen folder. Your website opens at http://localhost:8080/; the calculator shows S$693,138.
- **Lab folder:** labs/lab-01-studio-and-website/

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Lab 2 — Form the Team: Instructions for Each Agent · Slides 39–44

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (subagent files)
- **Time: 40 minutes (Day 1)**
- **Features practised:** five subagent files in .claude/agents/ · name, description, tools, model · the Lead is your session
- **Output:** .claude/agents/: growth-strategist, content-creator, creative-designer, website-designer, growth-analyst
- **Evidence:** Five files exist in .claude/agents/, one per specialist. You changed one instruction and can say why.
- **Lab folder:** labs/lab-02-agent-instructions/

Lab 3 — Give Each Agent Its Skills · Slides 47–52

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (.claude/skills/)
- **Time: 45 minutes (Day 1)**
- **Features practised:** twenty skills owned by role — research, writing, LinkedIn, Facebook, blog, newsletter, web design, lead magnet, video, analysis and compliance · SKILL.md · the "Use when" trigger · test without naming
- **Output:** twenty skills in .claude/skills/, wired to the agents that own them
- **Evidence:** Twenty folders in .claude/skills/, each with a SKILL.md. fin-compliance flagged "guaranteed 8%" as Critical.
- **Lab folder:** labs/lab-03-agent-skills/

Lab 4 — Connect the Tools and Check Compatibility · Slides 57–62

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude desktop app (Customize → Connectors), Claude Code
- **Time: 45 minutes (Day 1)**
- **Features practised:** Customize → Connectors: Google Drive · Gmail (drafts) · Google Calendar · Firecrawl · /plugin: Marketing (by Anthropic) · built-in browser · least privilege per agent · a compatibility matrix
- **Output:** every specialist tested on its own connector, with its limits written into its instructions, and a compatibility matrix for the whole team
- **Evidence:** /mcp lists Google Drive, Gmail, Google Calendar and Firecrawl; /plugin shows Marketing. The compatibility matrix gives every row a status, and a fix for anything not Compatible.
- **Lab folder:** labs/lab-04-agent-connectors/

Lab 5 — Campaign 1: Research to Storyboard, with Subagents · Slides 66–72

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (subagents in parallel)

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- **Time: 40 minutes (Day 1)**
- **Features practised:** the Lead delegates · strategist and analyst in parallel · nine scored ideas · a human concept checkpoint · a five-beat storyboard
- **Output:** competitor research, the funnel, nine scored ideas, the campaign brief and a five-beat storyboard with three channel variants — shown as artifacts, with a Word copy of each
- **Evidence:** The strategist and the analyst ran in parallel. You confirmed two claims; after Prompt C no High issue is left.
- **Lab folder:** labs/lab-05-campaign-1-research-to-storyboard/

Topic 2 — Testing Marketing Integrations and Troubleshooting Errors · Slides 74–138

- Integration tools, techniques and protocols
- Every campaign is an integration test
- Personas, cadence and a blog: test the data paths
- Social, newsletter, landing page and video: test the gate and the APIs
- Signs of integration errors, and basic troubleshooting

Lab 6 — Growth Analyst: Personas from Evidence · Slides 79–84

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (three analyst subagents in parallel)
- **Time: 40 minutes (Day 1)**
- **Features practised:** three analyst sub-agents, one per data file · counts, not impressions · confidence labels · content requirements
- **Output:** three evidence-based personas and the content spec — shown as artifacts, with a Word copy of each
- **Evidence:** Three analyst sub-agents ran, one per file. The writing guide answers every question for every persona.
- **Lab folder:** labs/lab-06-personas-from-evidence/

Lab 7 — Growth Analyst and Strategist: Cadence and Calendar · Slides 86–91

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (subagents)
- **Time: 35 minutes (Day 1)**
- **Features practised:** marginal value vs fatigue · team capacity · review buffers · a four-week calendar
- **Output:** how often to post on each channel, and a four-week calendar — shown as artifacts, with a Word copy of each
- **Evidence:** The cadence gives a frequency per channel with the evidence. No week is over 12 hours.
- **Lab folder:** labs/lab-07-cadence-and-calendar/

Lab 8 — Content Creator: A Quick Blog Article · Slides 93–97

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder

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- **Surface:** Claude Code (Firecrawl, your Horizon skills) → your copy of Horizon's website
- **Time: 40 minutes (Day 1)**
- **Features practised:** one prompt runs the whole job: research · draft · fact-check and fin-compliance · fix · post to the blog on your copy of the website · a person reads it last
- **Output:** a 500-700-word blog article with dated sources, checked and fixed and posted on your own copy of Horizon's website, opened in your browser, with a Word copy
- **Evidence:** Two sources, each with a date and a link. Your article is first under "From our blog" on your copy of the website.
- **Lab folder:** labs/lab-08-blog-article/

Lab 9 — The Lead Magnet: a Checklist and Its Welcome Emails · Slides 100–107

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (artifact, Gmail drafts, Marketing plugin, your Horizon skills)
- **Time: 50 minutes (Day 2)**
- **Features practised:** lead-magnet skill · a printable A4 checklist as an artifact · the Marketing plugin's email sequence · newsletter and sunny-voice · drafts only
- **Output:** the Money Check-up Checklist as a printable A4 page (artifact and PDF) and on your copy of Horizon's website, and three welcome emails — shown as artifacts, with a Word copy of each
- **Evidence:** The checklist uses Horizon's colours and all ten items. The checklist is on your copy of the website: email and consent first, then the ten tick boxes.
- **Lab folder:** labs/lab-09-lead-magnet/

Lab 10 — Campaign 2: Social Media Posts (Demo) · Slides 117–122

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (agent team) → LinkedIn, Facebook
- **Time: 60 minutes (Day 2)**
- **Features practised:** the publisher kit and its hook · an agent team: creator, designer, Lead · teammates message each other · human approval → you post
- **Output:** two posts with images, reviewed, approved by you and posted by you (or, optionally, through the API) — shown as artifacts, with a Word copy of each
- **Evidence:** The gate stopped Claude from approving. You posted the approved text and image yourself (or, optionally, through the API after you said go).
- **Lab folder:** labs/lab-10-campaign-2-social-week/

Lab 11 — Campaign 3: Newsletter and a Landing Page · Slides 125–130

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (agent team) → Claude artifact, Gmail
- **Time: 60 minutes (Day 2)**
- **Features practised:** creator writes, website designer builds the lead-magnet page, analyst filters by consent, the Lead checks · drafts only

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- **Output:** the November newsletter, its recipient list, the landing page published, and Gmail drafts — shown as artifacts, with a Word copy of each
- **Evidence:** Three specialists worked on the campaign; the Lead checked all. Drafts only — nothing was sent to the list.
- **Lab folder:** labs/lab-11-campaign-3-newsletter-and-landing-page/

Lab 12 — Campaign 4: The YouTube Explainer · Slides 132–137

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (agent team, render, upload)
- **Time: 45 minutes (Day 2)**
- **Features practised:** strategist angle · creator script · designer storyboard, thumbnail and render · approve twice · private upload
- **Output:** the script, storyboard and thumbnail, the video, and a private YouTube upload (or dry run) — shown as artifacts, with a Word copy of each
- **Evidence:** The strategist, creator and designer each delivered their part. The upload is private (or the dry run printed the request).
- **Lab folder:** labs/lab-12-campaign-4-youtube-explainer/

Topic 3 — Versioning, Change Proposals and Integration Best Practices · Slides 139–176

- Troubleshooting methodologies
- Always on: from skill to schedule
- Responsible AI; measure what matters
- Version the team; propose a change; roll back
- Agent teams at work: argue a strategy, then a QA loop

Lab 13 — Campaign 5: Always On — the Weekly Growth Report · Slides 144–149

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (skill, routine, Drive, Gmail)
- **Time: 45 minutes (Day 2)**
- **Features practised:** the Growth Analyst's campaign-report skill · do · verify · schedule · draft-only delivery
- **Output:** a scheduled task that drafts the weekly growth report every Monday at 8am, and one draft in Gmail
- **Evidence:** The report used the campaign-report skill from your folder. The task is paused after class.
- **Lab folder:** labs/lab-13-campaign-5-weekly-growth-report/

Lab 14 — Measure, Version and Write the Team Playbook · Slides 154–160

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (subagents)
- **Time: 60 minutes (Day 2)**

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- **Features practised:** compare agent patterns · the roster: instructions, skills, connectors · approval matrix · responsible-AI controls · version history · a change proposal and a roll-back plan
- **Output:** the agent-pattern comparison, the team playbook, a change proposal, and versions 1.0 and 1.1 of the team — shown as artifacts, with a Word copy of each
- **Evidence:** The comparison shows the one-agent baseline caught 0 defects. The change proposal has the evidence, the test, the risk and the roll-back.
- **Lab folder:** labs/lab-14-team-playbook/

Lab 15 — Agent Teams: A Strategy Debate and a QA Loop · Slides 164–170

- **Use:** Claude Code — Code mode (the </> button) in the Claude desktop app, in your horizon-studio folder
- **Surface:** Claude Code (agent teams: Lead, teammates, shared task list, direct messages)
- **Time: 35 minutes (Day 2)**
- **Features practised:** an advocate and a critic argue the pros and cons · the Lead judges · a writer and a QA Auditor loop until PASS · FAIL goes straight back to the writer · human approval
- **Output:** the decision (pros, cons, verdict), then an email, a video script and a thumbnail that passed QA, submitted for your approval — shown as artifacts, with a Word copy of each
- **Evidence:** The advocate and the critic messaged each other; every point cites a file. Both teams were shut down and cleaned up.
- **Lab folder:** labs/lab-15-agent-teams-debate-and-qa-loop/

Resources Required

- Trainer: the v3.0 slide deck, this Lesson Plan, the Learner Guide and the lab pack (labs/, 15 folders, with reference solutions for Labs 1, 2, 3, 10 and 12).
- Each learner: a laptop with the Claude desktop app on a Pro plan or above (Claude Code and artifacts are not on the Free plan), signed in with their claude.ai account. Learners type no commands: each lab says whether to use Claude or Claude Code.
- Installed by the trainer before class: Claude Code, Node.js 22, Python 3 with Pillow and ffmpeg (they run the approval gate and the video). A personal Google account (Drive, Gmail, Calendar). A free Firecrawl account (1,000 credits a month, no card) for web research; Anthropic's Marketing plugin (free, added in Lab 4).
- Lab 15 (agent teams) runs in the same studio folder; agent teams come switched on in .claude/settings.json.
- Optional, for live publishing: a LinkedIn developer app, a test Facebook Page and a Google Cloud project with the YouTube Data API. Every lab also works in dry-run mode without them.
- All fixtures are synthetic. Every learner email address resolves to the learner's own Gmail via plus-addressing.

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